MATH3900

Tutorial 5

1. Determine whether each statement is *True* or *False*. Give a short justification, or a counterexample with verification. You may use results proved in the lecture. Here P_C denotes Euclidean projection onto a nonempty, closed, convex set C.

   (a) For $C = \{x \in \mathbb{R}^2 : \|x\| \leq 1\}$, the vector $s = (0,1)^T$ is a feasible direction at $x = (1,0)^T$.

   (b) For every $u \in C$ and $v \in \mathbb{R}^d$,

   $$\|P_C(v) - u\| \leq \|v - u\|.$$

   (c) Let $f : \mathbb{R}^d \to \mathbb{R}$ be differentiable. If $x^\star \in C$ satisfies

   $$\nabla f(x^\star)^T (v - x^\star) \geq 0 \qquad \text{for every } v \in C,$$

   then $x^\star$ must be a global minimizer of f over C.

   (d) For every $u, v \in \mathbb{R}^d$, projection preserves their average:

   $$P_C\left(\frac{u+v}{2}\right) = \frac{P_C(u) + P_C(v)}{2}.$$

2. Consider

   $$\min_{x \in C} f(x), \qquad f(x) = \tfrac{1}{2}\big[(x_1 - 3)^2 + (x_2 - 1)^2\big],$$
   $$C = \{x \in \mathbb{R}^2 : x_1 \geq 0,\ x_2 \geq 0,\ x_1 + x_2 \leq 2\}.$$

   Let $x^\star = (2,0)^T$.

   (a) Write the constraints as $g_i(x) \leq 0$, $i = 1,2,3$, in the order displayed. Identify the active inequality set $\mathcal{A}(x^\star)$.

   (b) Find all feasible directions $s = (s_1, s_2)^T$ at $x^\star$. Justify both necessity and sufficiency using $x^\star + ts$.

   (c) Use the first-order optimality condition over a convex set to prove that $x^\star$ is the unique global minimizer. Is $\nabla f(x^\star) = 0$? Explain what happens to the ordinary gradient step $x^\star - \alpha \nabla f(x^\star)$ for any $\alpha > 0$.

3. Let

$$C = \{x \in \mathbb{R}^2 : x_1 + 2x_2 \leq 2\}, \qquad u = (2,2)^T, \qquad a = (1,2)^T.$$

Use the projection characterization from the lecture: for $z \in C$,

$$z = P_C(u) \quad \Longleftrightarrow \quad (u-z)^T(v-z) \leq 0 \quad \text{for every } v \in C.$$

(a) The point $\widehat{z} = (0,1)^T$ is feasible. Use $v = (1,1/2)^T$ in the characterization to show that $\widehat{z} \neq P_C(u)$.

(b) Look for a candidate of the form $z = u - ta$ on the boundary $z_1 + 2z_2 = 2$. Find t and z, then verify that $z = P_C(u)$ using the characterization for an arbitrary $v \in C$. Compute $\|u - z\|^2$.

(c) For the projection z found in (b), expand $\|u - v\|^2$ using $u - v = (u - z) - (v - z)$. Show that

$$\|u - v\|^2 \geq \frac{16}{5} + \|v - z\|^2 \qquad \text{for every } v \in C.$$

Explain why no other feasible point is as close to u as z.

4. Use the same halfspace C, point u, and projection $z = P_C(u)$ as in Question 3. Consider two perturbations of the input:

$$d_T = (1/5, -1/10)^T, \qquad d_N = (1/10, 1/5)^T, \qquad u_T = u + d_T, \quad u_N = u + d_N.$$

(a) Calculate $\|d_T\|$ and $\|d_N\|$. Without computing the new projections, use nonexpansiveness to bound $\|P_C(u_T) - z\|$ and $\|P_C(u_N) - z\|$.

(b) Verify, using the projection characterization, that

$$P_C(u_T) = z + d_T = (7/5, 3/10)^T, \qquad P_C(u_N) = z = (6/5, 2/5)^T.$$

(c) Compute the actual changes in the two projections. For which perturbation is the bound in (a) attained? Explain the difference geometrically using the boundary $x_1 + 2x_2 = 2$ and its normal vector $a = (1, 2)^T$.